Medicine

Total Intravenous Anesthesia and Target-Controlled Infusion in Pediatric Minimally Invasive Cardiac Surgery: A Single-Center Retrospective Cohort Study

--Manuscript Draft--

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Corresponding Author:	Pinar Ozdemir Yasar Hacettepe University: Hacettepe Universitesi Ankara, Cankaya TURKEY
First Author:	Pınar Özdemir Yaşar, MD
Order of Authors:	Pınar Özdemir Yaşar, MD
	Başak Akça, MD
	Şafak Alpat, MD
	Mustafa Yılmaz, MD
Manuscript Region of Origin:	TURKEY
Abstract:	Background : While minimally invasive cardiac surgery has gained growing acceptability in adults, expertise in young patients—particularly regarding anesthetic management—remains scarce. This study evaluated anesthetic methods, perioperative results, and safety profiles in pediatric patients undergoing minimally invasive cardiac surgery at a single tertiary care facility. Methods : This single-center retrospective cohort study included all pediatric patients aged 5 to 16 years (ASA physical status I to III) who underwent elective minimally invasive cardiac surgery (MICS) through a right anterior mini-thoracotomy for atrial septal defect or mitral regurgitation between 2015 and 2025 (n=8). Anesthesia was delivered with propofol and remifentanil as either total intravenous anesthesia (TIVA, n=5) or target-controlled infusion (TCI, n=3). The primary endpoints were cross-clamp time, time to extubation in the intensive care unit (ICU), ICU and total hospital length of stay, vasoactive-inotropic requirement and 30-day mortality; the secondary endpoint was the occurrence of major postoperative complications. Data were summarized descriptively, proportions are given with exact 95% confidence intervals (CIs), and no comparative hypothesis testing was performed. Results : Median surgical duration was 210 mins (185–240 mins), anesthesia duration 235 mins (220–260 mins), cardiopulmonary bypass duration 101 mins (91.5–106.5 mins), and cross-clamp duration 63 mins (50–70 mins). Median extubation duration was 150 mins (125–220 mins), median ICU length of stay was 24 hrs (23.5–25.5 hrs), total hospital length of stay was 6.0 [5.0–8.0] days for the TIVA group and 8.0 [6.5–8.5] days for the TCI group, and 30-day mortality was 0% (0/8; 95% CI 0–36.9%). Inotropic support was required in 4 of 8 patients (50%; 95% CI 15.7–84.3%)and successfully discontinued within the first 12 hours postoperatively. The secondary endpoint of postoperative major complications was 0% (95% CI 0–36.9%). Conclusions : This small single-center retrospective cohort study indicates that pediatric minimally invasive cardiac surgery managed with TIVA or TCI is associated with favorable preliminary perioperative outcomes and stable intraoperative courses in selected patients. Given the exploratory design and limited cohort size, these findings should be interpreted with caution. Nonetheless, this initial experience contributes to expanding clinical literature and underscores the role of multidisciplinary expertise in optimizing patient safety and care.

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24 October 2026

RE: MD-D-26-05075

Previous title: "Total Intravenous Anesthesia Versus Target-Controlled Infusion in Pediatric Minimally Invasive Cardiac Surgery: A Retrospective Study"

Revised title: "Total Intravenous Anesthesia and Target-Controlled Infusion in Pediatric Minimally Invasive Cardiac Surgery: A Single-Center Retrospective Cohort Study"

Dear Editor-in-Chief,

Medicine

We thank you, Reviewer #2 and Reviewer #3 for their further assessment of our manuscript. We are grateful to Reviewer #2 for recommending publication. Reviewer #3 raised a single substantive point, the classification of the study design, which we have considered carefully. For the reasons set out below we respectfully maintain that the work is a retrospective cohort study, and we have revised the manuscript so that its design, its study size and the precision of its estimates are stated explicitly and consistently throughout.

All new and altered text appears in yellow in the revised manuscript file. Our point-by-point responses follow.

Summary of the principal changes

1. Title. The design is now stated as a single-center retrospective cohort study, and “Versus” has been replaced by “and”, since the study describes both techniques and does not compare them.

2. Methods, design statement. A paragraph now defines the design: patients were identified by exposure (elective MICS under one of two predefined intravenous techniques) within fixed eligibility criteria and a fixed period and were followed for

predefined outcomes. The study is reported according to the STROBE statement, which replaces the reference to the CARE guideline for case reports and case series.

3. Methods, new Study Size subsection. The subsection explains that study size was set by all eligible patients treated during the study period, why an a priori power calculation does not apply to a descriptive study, and why no post hoc power calculation was performed.

4. Precision of estimates. Exact (Clopper–Pearson) 95% confidence intervals have been added for the key proportions in the Abstract and Results: 30-day mortality and major complications 0/8 (95% CI 0–36.9%), and inotropic support 4/8 (50%; 95% CI 15.7–84.3%).

5. Abstract. The case-report headings (Rationale, Patient concerns, Diagnosis, Interventions, Outcomes, Lessons) have been replaced by Background, Methods, Results and Conclusions. The “Interventions” heading has been removed.

6. Terminology. Every description of the work as a “case series” has been replaced, in the Abstract, Methods, Discussion and Conclusions.

7. Limitations. The Limitations now state that the study size was not based on a sample size calculation, that the study was not powered to compare the two techniques or to detect uncommon complications, and how wide the confidence interval around a complication rate of 0% remains.

8. References and abbreviations. Three methodological references have been added as references 8 to 10 (the STROBE statement, its explanation and elaboration document, and Dekkers et al. on distinguishing case series from cohort studies). The former references 8 to 25 have been renumbered 11 to 28 in order of citation. The list of abbreviations has been completed and a typographical error (CBP) corrected to CPB.

Reviewer #2

Comment. “*As I said, the revision is good enough to be published.*”

Response. We thank the reviewer for the careful assessment over both rounds of review and for recommending publication. No further changes were requested.

Reviewer #3

Q1. "*The papers is best to be a case series, as retrospective study needs sample size and power calculation.*"

Response. We thank the reviewer for raising this point, which led us to make the design, study size and precision of the work explicit in the manuscript. We have considered the suggestion carefully and respectfully maintain that the study is correctly classified as a retrospective cohort study. Our reasons are as follows.

- **Design is defined by how patients are sampled, not by their number.** In the methodological framework set out by Dekkers et al. (Ann Intern Med 2012; reference 9), a cohort study samples patients on exposure and follows them for outcomes, and it can do so without a comparison group, whereas a case series samples patients on the basis of a disease or outcome. Our patients were identified by exposure (elective MICS through right anterior mini-thoracotomy under TIVA or TCI) within predefined eligibility criteria and a fixed ten-year window, and all eligible patients were followed for predefined outcomes (time to extubation, intensive care and hospital stay, vasoactive requirement, 30-day mortality and major complications). This is the structure of a retrospective cohort, and it allows absolute event frequencies to be estimated, which a case series cannot do.

- **A power calculation applies to hypothesis tests, and this study tests none.** Sample size and power calculations exist to ensure that a planned comparison can detect a prespecified effect. After the previous round of review we removed every comparative test and p-value; the study has a descriptive objective only. For retrospective studies in which the study size is fixed by the available population, the STROBE explanation and elaboration document (reference 10, item 10) asks authors to explain how the study size was arrived at, not to report an a priori power calculation, and it advises explicitly against post hoc power calculations, recommending confidence intervals instead. We have followed this guidance.

- **The precision of the estimates is now stated.** In place of a power calculation, the Abstract and Results now give exact 95% confidence intervals for the key proportions. For example, the observed 30-day mortality and major complication rate of 0/8 carries an upper confidence limit of 36.9%. This conveys the limited informativeness of eight patients more directly than a power figure would, and the Limitations now state it in those terms.

- **The small size of the cohort is acknowledged, not concealed.** We agree with the reviewer that eight patients cannot support comparative inference. The manuscript states this in the Methods, Results, Discussion and Limitations, and draws no conclusion about the relative merits of TIVA and TCI.

The corresponding changes are:

- **Methods, opening paragraph:** The design is stated as a single-center retrospective cohort study reported according to STROBE (reference 8), with the sampling logic that defines it (reference 9). The reference to the CARE guideline for case reports and case series has been removed.

- **Methods, new Study Size subsection:** “The study size was determined by the total number of eligible patients treated at our institution during the study period, not by an a priori calculation. Because the objective was descriptive and no hypothesis was tested, a formal power calculation was not applicable. In line with the STROBE recommendations, no post hoc power calculation was performed; the precision of the estimates is instead conveyed by 95% CIs.”

- **Methods, Cohort:** All consecutive eligible patients were included, and 30-day follow-up was complete.

- **Statistical analysis, Abstract and Results:** Exact (Clopper–Pearson) 95% confidence intervals are now reported for proportions.

- **Limitations:** Absence of an a priori sample size calculation and the resulting lack of power for comparison or for detecting uncommon complications are stated explicitly.

- **Terminology:** “Case series” has been replaced throughout, and the title now names the design as a single-center retrospective cohort study.

We hope that these changes address the reviewer's concern by making the design and its limits transparent, while describing the study accurately in the terms used by the reporting guideline that governs it.

Editorial Formatting Comments

-Title page: Be sure the title page lists all author names, degrees, and affiliations.

-Title: Be sure the title includes any specific terms as directed in the reporting guidelines for your type of article (for example, "case report" should be in the title of a CARE-compliant article). The following guidelines specify that the study design should be in the article title: CARE, CHEERS, CONSORT, PRISMA, STARD, and STROBE.

-Abstract: Be sure to use a structured abstract, with headings. Use the specific headings listed in the guidelines checklist if your report is based on the CARE, CHEERS, CONSORT, or PRISMA guidelines.

-List of abbreviations: If not already included, please include a list of abbreviations used in the text as part of the manuscript text file following the title page. Use abbreviations sparingly in the text, and spell them out the first time you use them. Abbreviations used in tables should be spelled out at the bottom of the table.

-Ethical review, Methods section: If not already included, please state in the Methods section that an ethics committee or institutional review board approved the study, and list the board's name. If ethical approval was waived or not necessary, please state the reason. If the study involves patient consent, state explicitly that informed consent was or was not given, and state the reason if not given.

-Funding/Conflict of Interest information: List any source of funding or anything that could be perceived as a conflict of interest in the Acknowledgments section.

-Acknowledgments: If you list anyone by name in the Acknowledgments section, please confirm that the person gives permission to be named.

Response to the editorial formatting comments

We thank the editorial office for the checklist and have worked through it in order.

Title page. All author names, academic degrees and institutional affiliations are listed on the title page.

Title. In accordance with the STROBE statement, the study design (a single-center retrospective cohort study) is stated in the title.

Abstract. The abstract is structured under the headings Background, Methods, Results and Conclusions.

List of abbreviations. The list of abbreviations following the title page has been completed. Abbreviations are spelled out at first use in the text and are defined at the foot of each table.

Ethical review. The Methods state that the study was approved by the Institutional Review Board of Hacettepe University (SBA 2025/17-25, 09.09.2025) and that written informed consent was obtained from patients or their legal guardians.

Funding and conflicts of interest. The declarations state that the study received no specific grant from any funding agency and that the authors have no conflicts of interest to declare.

Acknowledgments. No individuals are named in the Acknowledgments, so no permissions were required.

We hope that the revised manuscript now meets the standards of Medicine and we look forward to your decision.

27.10.2026

Dear Editor / Editorial Office,

Thank you for reviewing our manuscript entitled **"Total Intravenous Anesthesia and Target-Controlled Infusion in Pediatric Minimally Invasive Cardiac Surgery: A Single-Center Retrospective Cohort Study"** (Ref No: MD-D-26-05075R2).

We received the technical check feedback requesting us to format the abstract using Case Report headings (*Rationale, Patient concerns, Diagnosis, Interventions, Outcomes, Lessons*) and to add "Case Report" to the manuscript title.

We would like to humbly clarify that our manuscript is not an individual case report or case series, but a **single-center retrospective observational cohort study** involving a patient cohort evaluated according to exposure groups. Consequently, the study design and manuscript structure strictly adhere to the **STROBE (Strengthening the Reporting of Observational Studies in Epidemiology)** guidelines, utilizing standard observational study headings (*Background, Methods, Results, Conclusions*).

Applying Case Report guidelines (*CARE*) to a retrospective cohort study might misrepresent the study design and statistical methodology to the readership.

Therefore, we kindly request your guidance on whether we may retain the current observational cohort format (*Background, Methods, Results, Conclusions*) and title, or if the journal explicitly requires us to reformat the manuscript under the Case Report framework despite its cohort study methodology.

We appreciate your time, understanding, and continued support during the editorial process.

Sincerely,

Dr. Pınar Özdemir Yaşar

Corresponding Author

Department of Anesthesiology and Reanimation

Hacettepe University Faculty of Medicine

OA_Manuscript

Total Intravenous Anesthesia and Target-Controlled Infusion in Pediatric Minimally Invasive Cardiac Surgery: A Single-Center Retrospective Cohort Study

Author names

Pınar Özdemir Yaşar, MD[1], Başak Akça MD[1], Şafak Alpat MD[2], Mustafa Yılmaz MD[2]

Affiliations

[1]Hacettepe University Faculty of Medicine Hospital, Department of Anesthesiology and Reanimation, Ankara, Turkey

[2]Hacettepe University Faculty of Medicine Hospital, Department of Cardiovascular Surgery, Ankara, Turkey

List of Abbreviations

ASA: American Society of Anesthesiologists
ASD: Atrial Septal Defect
BIS: Bispectral Index
BMI: Body Mass Index
BSA: Body Surface Area
CI: Confidence Interval
CPB: Cardiopulmonary Bypass
DSA: Density Spectral Array
EEG: Electroencephalography
FiO_2: Fraction of Inspired Oxygen
HFNC: High-Flow Nasal Cannula
ICU: Intensive Care Unit
IQR: Interquartile Range
MICS: Minimally Invasive Cardiac Surgery
MR: Mitral Regurgitation
NIRS: Near-Infrared Spectroscopy
OLV: One-Lung Ventilation
PEEP: Positive End-Expiratory Pressure
STROBE: Strengthening the Reporting of Observational Studies in Epidemiology
TCI: Target-Controlled Infusion
TIVA: Total Intravenous Anesthesia
VSD: Ventricular Septal Defect

Corresponding Author

Name: Dr. Pınar Özdemir Yaşar

Address: Hacettepe University Faculty of Medicine Department of Anesthesiology,
Sıhhiye, Ankara, 06100, TURKEY
Phone: +903123051255
Fax: +903123109600
E-Mail: pozdemiryasar@gmail.com

Abstract

Background: While minimally invasive cardiac surgery has gained growing acceptability in adults, expertise in young patients—particularly regarding anesthetic management—remains scarce. This study evaluated anesthetic methods, perioperative results, and safety profiles in pediatric patients undergoing minimally invasive cardiac surgery at a single tertiary care facility.

Methods: This single-center retrospective cohort study included all pediatric patients aged 5 to 16 years (ASA physical status I to III) who underwent elective minimally invasive cardiac surgery (MICS) through a right anterior mini-thoracotomy for atrial septal defect or mitral regurgitation between 2015 and 2025 (n=8). Anesthesia was delivered with propofol and remifentanil as either total intravenous anesthesia (TIVA, n=5) or target-controlled infusion (TCI, n=3). The primary endpoints were cross-clamp time, time to extubation in the intensive care unit (ICU), ICU and total hospital length of stay, vasoactive-inotropic requirement and 30-day mortality; the secondary endpoint was the occurrence of major postoperative complications. Data were summarized descriptively, proportions are given with exact 95% confidence intervals (CIs), and no comparative hypothesis testing was performed.

Results: Median surgical duration was 210 mins (185–240 mins), anesthesia duration 235 mins (220–260 mins), cardiopulmonary bypass duration 101 mins (91.5–106.5 mins), and cross-clamp duration 63 mins (50–70 mins). Median extubation duration was 150 mins (125–220 mins), median ICU length of stay was 24 hrs (23.5–25.5 hrs), total hospital length of stay was 6.0 [5.0–8.0] days for the TIVA group and 8.0 [6.5–8.5] days for the TCI group, and 30-day mortality was 0% (0/8; 95% CI 0–36.9%). Inotropic support was required in 4 of 8 patients (50%; 95% CI 15.7–84.3%) and successfully discontinued within the first 12 hours postoperatively. The secondary endpoint of postoperative major complications was 0% (95% CI 0–36.9%).

Conclusions: This small single-center retrospective cohort study indicates that pediatric minimally invasive cardiac surgery managed with TIVA or TCI is associated with favorable preliminary perioperative outcomes and stable intraoperative courses in selected patients. Given the exploratory design and limited cohort size, these findings should be interpreted with caution. Nonetheless, this initial experience contributes to expanding clinical literature and underscores the role of multidisciplinary expertise in optimizing patient safety and care.

Keywords: MICS, TIVA, TCI, Pediatric cardiac surgery, Retrospective studies

What is already known?

Minimally invasive cardiac surgery (MICS) in pediatric patients is a domain where advances in perioperative management intersect with evolving anesthesia techniques.

The feasibility of performing MICS in children depends heavily on accurate patient selection and the availability of coordinated expertise from surgical, anesthetic, and critical care teams.

What this article adds

This study confirms that adult MICS protocols are transferable to pediatric care through developmental adjustments and multimodal approaches like TIVA/TCI. It highlights how the synergy between established adult practices and pediatric-specific innovations can yield reproducible positive outcomes and rapid recovery without compromising procedural safety or long-term wellbeing.

INTRODUCTION

Minimally invasive cardiac surgery (MICS) has transformed the treatment of congenital and acquired heart disorders by diminishing surgical trauma, improving aesthetic outcomes, and promoting expedited recovery relative to traditional sternotomy [1, 2]. Although these methods are becoming more prevalent among adults, their use in the pediatric demographic has been constrained, mostly due to technical difficulties, reduced patient size, and safety concerns [3, 4].

Numerous studies have shown that minimally invasive techniques, such as right anterior mini-thoracotomy, axillary thoracotomy, and partial sternotomy, can yield outstanding results in pediatric patients with uncomplicated congenital heart anomalies, including atrial septal defects (ASD) and ventricular septal defects (VSD) [5, 6]. Moreover, longitudinal follow-up data validate the longevity and aesthetic advantages of these procedures [7]. Notwithstanding these advancements in surgical practice, data on anesthetic administration during pediatric minimally invasive cardiac surgery is limited.

Minimally invasive cardiac surgery (MICS) in pediatric patients is a domain where advances in perioperative management intersect with evolving anesthesia techniques. Anesthetic concerns in pediatric minimally invasive cardiac surgery (MICS) are distinctive, encompassing problems related to vascular access, cardiopulmonary bypass techniques, analgesia, and early extubation protocols [4, 5]. Methods like total intravenous anesthesia (TIVA) and target-controlled infusion (TCI) are progressively utilized in adult minimally invasive cardiac surgery (MICS), although their effectiveness and outcomes in pediatric patients remain inadequately defined [3]. The feasibility of performing MICS in children depends heavily on accurate patient selection and the availability of coordinated expertise from surgical, anesthetic, and critical care teams.

This study confirms that adult MICS protocols are transferable to pediatric care through developmental adjustments and multimodal approaches like TIVA/TCI. It highlights how the synergy between established adult practices and pediatric-specific innovations can yield reproducible positive outcomes and rapid recovery without compromising procedural safety or long-term wellbeing. The present retrospective cohort study describes our unit's experience with pediatric patients undergoing MICS with a focus on anesthetic management and perioperative results.

MATERIALS AND METHODS

This retrospective study was conducted in the operating theaters of the Department of Anesthesiology and Reanimation at Hacettepe University between 2015 and 2025. Ethical approval was granted by the Institutional Review Board of Hacettepe University (Approval No: SBA 2025/17-25, Date: 09.09.2025), and written informed consent was obtained from all participating patients or their legal guardians. The study was designed as a single-center retrospective cohort study and is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement [8]. Patients were identified by their exposure, namely elective MICS under

one of two predefined intravenous anesthetic techniques, within fixed eligibility criteria and a fixed calendar period, and were then followed through their records for predefined perioperative outcomes; sampling on exposure rather than on outcome defines a cohort rather than a case series, irrespective of cohort size or the presence of a comparison group [9]. The study cohort included pediatric patients aged 5 to 16 years, with an American Society of Anesthesiologists (ASA) physical status of I to III, who were scheduled for elective minimally invasive cardiac surgery. Specifically, the procedures were restricted to right anterior mini-thoracotomy approaches for the surgical correction of Atrial Septal Defect (ASD) and Mitral Regurgitation (MR). Patients requiring emergency surgical interventions were excluded from the study.

Cohort

All pediatric patients who underwent MICS at our institution between 2015 and 2025 and met the eligibility criteria were included consecutively, giving a cohort of 8 patients. We retrospectively collected relevant demographic data (age, weight, BMI, BSA, ASA classification), disease characteristics, intraoperative data (total surgical and anesthetic durations, CPB and cross-clamp times, fluid balance) and postoperative factors, such as vasoactive inotrope scores, rhythm traces, time to extubation, and length of stay in critical care and in hospital from patients. Follow-up to 30 days was complete in all patients.

Anesthetic Technique

Upon arrival in the operating theater, standard cardiac anesthetic monitoring was established for all patients. This included five-lead electrocardiography, pulse oximetry, capnography, invasive arterial blood pressure, central venous pressure (CVP), near-infrared spectroscopy (NIRS), and processed EEG via Bispectral Index (BIS) to assess the depth of anesthesia. External defibrillator pads were positioned prior to draping to facilitate a prompt response in the event of arrhythmias or fibrillation.

General anesthesia was induced utilizing either target-controlled infusion (TCI) or conventional total intravenous anesthesia (TIVA) with propofol and remifentanil. In the TCI group (n=3), induction was achieved using the Eleveld pharmacokinetic model for propofol, set to an effect-site concentration (Ce) of 4.0 µg/mL, and the Minto pharmacokinetic model for remifentanil, set to a Ce of 4.0 ng/mL. In the TIVA group (n=5), induction was initiated with continuous infusions of propofol at a rate of 8 mg/kg/h and remifentanil at 0.05 µg/kg/min. Following the loss of consciousness, neuromuscular blockade was administered using rocuronium (0.6 mg/kg) to facilitate endotracheal intubation, with the depth of the blockade continuously monitored. Following intubation, one-lung ventilation (OLV) was implemented for all patients. Mechanical ventilation parameters included a tidal volume of 5–6 mL/kg, a positive end-expiratory pressure (PEEP) of 5 cmH_2O, and an inspired oxygen fraction (FiO_2) of 0.5–0.8, dynamically adjusted to maintain oxygen saturation >92%. During the maintenance of anesthesia, the administration of both agents was continuously titrated guided by the Density Spectral Array (DSA) trends of the processed EEG to ensure an optimal

depth of anesthesia. In the TCI group, the effect-site concentrations (Ce) were dynamically adjusted using the aforementioned pharmacokinetic models, typically ranging between 3.0 and 4.0 µg/mL for propofol and 2.0 to 4.0 ng/mL for remifentanil, in response to varying surgical stimuli and continuous DSA monitoring. In the TIVA group, adequate anesthesia depth was maintained by titrating continuous infusions of propofol at 6–10 mg/kg/h and remifentanil at 0.1–0.3 µg/kg/min according to the DSA patterns. Effective analgesia was sustained throughout the surgical procedure via the continuous administration of remifentanil in both modalities

All patients underwent a right anterior mini thoracotomy. Peripheral cardiopulmonary bypass (CPB) was established through femoral cannulation. Myocardial protection was attained with antegrade cold blood cardioplegia subsequent to the application of the aortic cross-clamp. Intravenous fluid management consisted of balanced crystalloids at a restrictive rate of 3–5 mL/kg/h, and hemodynamic stability was maintained using boluses of ephedrine (0.1–0.2 mg/kg) or a continuous infusion of norepinephrine (0.02–0.05 µg/kg/min) if the mean arterial pressure decreased by more than 20% from the patient's baseline.

Given the descriptive objective of this retrospective cohort study and its limited sample size (n=8), formal comparative statistical analyses between the TIVA and TCI groups were not performed. Instead, our outcomes were strictly defined as primary and secondary observational endpoints. The primary observational endpoints included cross-clamp time, postoperative extubation time in the intensive care unit (ICU), ICU length of stay, total hospital length of stay, vasoactive-inotropic requirements, and 30-day mortality. The secondary observational endpoint was defined as the occurrence of postoperative major complications.

Study Size

The study size was determined by the total number of eligible patients treated at our institution during the study period, not by an a priori calculation. Because the objective was descriptive and no hypothesis was tested, a formal power calculation was not applicable. In line with the STROBE recommendations, no post hoc power calculation was performed; the precision of the estimates is instead conveyed by 95% CIs [8, 10].

Statistical analysis was performed using SPSS version 25.0 (IBM Corp, 2017). Due to the descriptive design and small cohort size (n=8), comparative analyses (such as hypothesis testing between TIVA and TCI groups) were intentionally omitted. All data are presented using descriptive statistics. Continuous variables are expressed as mean ± standard deviation (SD) or median with interquartile range (IQR), depending on data normality, and categorical variables are reported as frequencies and percentages (%). Proportions are reported with exact (Clopper–Pearson) 95% CIs.

RESULTS

Eight patients underwent MICS 4 of them were male (50 %) and 4 were female (50%). At the time of MICS, the median age and weight were 12 years (9 - 15 years) and 46 kg (37 - 52 kg), respectively. BMI and BSA 20.3 (10.7-

25.1), ; 1.4135 (0.573-1.794) respectively. The diagnoses for surgery were Atrial Septal Defect (ASD) and Mitral Regurgitation (MR). Table 1 summarizes the demographic and clinical characteristics of the patients.

Five patients received TIVA and three received TCI. The median surgical duration was 210 mins (185 –240 mins), anesthesia duration 235 mins (220 – 260 mins), CPB duration 101 mins (91.5 – 106.5 mins), and cross-clamp time 63 mins (50 – 70 mins). Total fluids and urine output 900 (530-1500) and 1000.0 (500.0-1500.0) respectively. The details of intraoperative management are shown in Table 2.

In all patients, the postoperative course was uncomplicated. 4 (50%; 95% CI 15.7–84.3%) patients required inotropic support, with a median vasoactive inotropic score of 1.25 (0.0–5.0), and all inotropes were discontinued by the 12th hour after the operation. There was no evidence of atrial or ventricular arrhythmia in the early postoperative period. We were able to extubate all patients in a median duration of 150 mins (125 - 220 mins), the median length of stay in the ICU was 24 hrs (23.5 – 25.5 hrs), and total hospital length of stay of 6.0 (5.0–8.0) days for the TIVA group and 8.0 (6.5–8.5) days for the TCI group. There was no 30-day mortality and no major complication (0/8 for each; 95% CI 0–36.9%). Postoperative data was depicted in Table 3.

Due to the descriptive design and small cohort size, formal comparative hypothesis testing was not performed; however, individual observations between the TIVA and TCI groups are presented to illustrate our clinical experience. Table 4 shows the results in detail.

DISCUSSION

Eight children underwent right anterior mini thoracotomy for atrial septal defect or mitral regurgitation over a ten-year period. Although a cohort of eight patients cannot test a technique, it can still record what was done and what followed.

Our experience indicates that the anesthetic requirements of adult minimally invasive cardiac surgery, which include one-lung ventilation, peripheral femoral bypass, effect-site targeted intravenous anesthesia, processed EEG and cerebral oximetry, can be met in a group of child patients of median weight 46 kg, and that recovery was not delayed to a degree that would favour sternotomy. While pediatric surgical studies have established the feasibility and durability of this approach [4–7], the specific details of anesthetic management are less frequently reported in the literature.

The duration of recovery in our current study aligns closely with similar timeframes previously recorded in the adult MICS research literature. White and colleagues describe patients extubated within a few hours and an intensive care stay of 24 to 48 hours [11], and the institutional protocol of Demir and Mavioğlu sets early extubation within four to six hours as the target in hemodynamically stable patients [12]. Our median of 150 minutes to extubation and 24 hours

in intensive care is therefore similar with adult ultra-fast-track standards, and the useful observation is that these standards were also met in children.

Jiang et all found that the oxygenation index falls immediately after ultra-fast-track extubation and recovers only by 12 hours, with more frequent recourse to high-flow nasal oxygen [13], so early extubation in a child requires a plan for respiratory support as well as a target time. In line with this, high-flow nasal cannula (HFNC) therapy was employed in our series to provide effective respiratory support whenever postoperative respiratory support was indicated.

Since there are five patients in TIVA and three patients in TCI group, this study cannot support any conclusion comparing the two techniques, and so no comparison was attempted. Total hospital stay was numerically longer in the TCI group (8.0 versus 6.0 days). The pediatric literature addresses this comparison without settling it for cardiac surgery: target-controlled and manual propofol delivery have been compared directly in children [14], as has bispectral index behaviour under the two modes [15], and intravenous maintenance has a long descriptive record in pediatric cardiac anesthesia [16]. Evidence from adults is more substantial. Bai et al. matched elderly cardiac surgical patients receiving target-controlled against manually controlled infusion across two international cohorts and found shorter intensive care stay, lower odds of reintubation, reduced blood pressure variability and a smaller burden of hypotension with TCI [17]. Their work differs from ours in two respects: their infusions ran on the Marsh and Minto models in elderly patients undergoing conventional surgery, whereas we used the latest TCI model “Eleveld” for propofol in children undergoing MICS. Our own method carries a further caveat. The remifentanil component was targeted with a model derived in adults, so its predicted effect-site concentrations in children of this age represent an operational setpoint, since actual exposure was not measured. In both arms the infusion was titrated against the density spectral array of the processed EEG, so the two groups differed in the interface used to reach a target while depth of anesthesia was monitored in the same way.

MICS anesthesia combines cardiac anesthesia with elements of thoracic practice, and most of what it adds concerns the airway [18]. Surgical access through a right anterior mini-thoracotomy requires lung isolation [19, 20], achieved either with a double-lumen tube or with a bronchial blocker whose position is confirmed bronchoscopically [20], and a left-sided double-lumen tube is the conventional choice [13]. In our cohort, lung isolation was successfully achieved in all patients using a left-sided double-lumen tube. External defibrillator pads are positioned before draping because internal cardioversion is often not possible during these procedures [20], and this was our routine in every case. Device dimensions, more than any anesthetic principle, limit how far these techniques transfer to a child. Our cohort, with a median age of 12 years and an interquartile range of 9 to 15, sat in the upper part of the pediatric range where adult-sized equipment becomes practicable.

Peripheral cannulation creates a monitoring problem that full sternotomy does not. When femoral arterial return meets blood ejected by a heart that is still contracting, the upper body can be perfused with deoxygenated blood, and cerebral

oximetry is often where this first becomes apparent [12, 20]. Aquilina et al. documented the sequence during minimally invasive mitral repair: right upper limb saturation fell to 65% after the venous cannula migrated out of the superior vena cava and returned to 100% once it was repositioned [21]. Demir and Mavioğlu give the operational thresholds, a regional saturation below 45 or a fall of more than 20% from baseline, and they place cannula position among the first things to check when these thresholds are breached [12]. The Bonn protocol also includes near-infrared spectroscopy and processed EEG in its standard monitoring bundle [22], and we used these monitors both in all of our patients. Cannulaes are smaller and circulating volüme is lower in a child, so the same degree of displacement can be expected to produce a larger fractional change in venous drainage, which is an argument for continuous cerebral oximetry in this population.

Analgesia is the part of our protocol with the clearest room for improvement. Maintenance analgesia was remifentanil in both arms, and it is well known that remifentanil provides no residual analgesia once the infusion stops. Marianello et al. administered a single-shot deep serratus anterior plane block an hour before extubation in adults undergoing minimally invasive mitral surgery and reduced the proportion of patients requiring rescue opioids over the first 24 hours from 84% to 20%, with better sedation scores [23]. Regional blockade also features in the ultra-fast-track data, where fascial plane blockade was an independent positive predictor of achieving ultra-fast-track management and intraoperative opioid dose a negative one [24]; higher intraoperative morphine equivalents likewise reduced the odds of operating room extubation in the series of Wang et al. [25]. A regional adjunct is therefore the addition most clearly indicated for a pediatric pathway of this kind.

Eight patients over ten years is roughly one per year. That volume reflects careful case selection, and it also limits what a single centre can demonstrate. These operations are offered to selected children by a team that performs them regularly in adults, and the scientific statement on percutaneous and minimally invasive valve procedures identified the learning curve and cross-specialty team training as determinants of outcome well before these techniques became routine [26]. Case reports continue to show the boundary moving where experience is concentrated, among them a mitral valve replacement through a mini-thoracotomy in a patient with a giant left atrium [27], and the anesthetic skill set has been described as a distinct competency within cardiac practice [28]. Almost all of the literature we have relied on here was written for adults, and until pediatric-specific data accumulate, careful extrapolation remains the only available route and should be identified as such. A multicentre registry recording the endpoints used in this report, time to extubation, intensive care stay, vasoactive requirement and oxygenation after extubation, would allow single-centre experiences of this kind to be pooled.

This small single-center retrospective cohort study indicates that pediatric MICS managed with either TIVA or TCI is associated with favorable preliminary perioperative outcomes in selected patients. Both anesthetic techniques exhibited stable intraoperative courses and uneventful early recovery profiles. Given the exploratory design and limited cohort size of this study, these findings should be interpreted with caution. Nonetheless, our initial experience

contributes to the expanding clinical data on minimally invasive techniques in pediatric cardiac surgery and underscores the value of multidisciplinary expertise in optimizing patient care.

The limitations of this study include its retrospective and observational nature in a tertiary referral center and the small number of subjects. The study size was set by the number of eligible patients and not by an a priori sample size calculation, so the study was not powered to compare the two anesthetic techniques or to detect uncommon complications; the upper 95% confidence limit of 36.9% for zero events among eight patients illustrates how wide the uncertainty around a complication rate of 0% remains. A notable limitation of this study is the specific scope of the surgical procedures analyzed. Although categorized under minimally invasive cardiac surgery (MICS), it is important to acknowledge that MICS encompasses a heterogeneous group of procedures, each presenting unique anesthetic implications and physiological challenges. Because our cohort consisted entirely of patients undergoing right anterior mini-thoracotomy for ASD and MR, our findings regarding anesthetic management and postoperative outcomes may not be entirely generalizable to other MICS approaches, such as mini-sternotomy or robotic-assisted cardiac surgeries. Furthermore, the lack of long-term follow-up data prevents inferences about late outcomes, including neurodevelopment and thoracic growth. The pharmacokinetic models used for target-controlled delivery are a further limitation. Remifentanil was targeted with the Minto model, which was derived in adults, so the effect-site concentrations recorded in the TCI group represent an operational setpoint that has not been validated in children. Further prospective multicenter research may help overcome these limitations.

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CONFLICTS OF INTEREST

None.

COMPETING INTERESTS

The author(s) declare that none

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Total Intravenous Anesthesia and Target-Controlled Infusion in Pediatric Minimally Invasive Cardiac Surgery: A Single-Center Retrospective Cohort Study

Author names

Pınar Özdemir Yaşar, MD[1], Başak Akça MD[1], Şafak Alpat MD[2], Mustafa Yılmaz MD[2]

Affiliations

[1]Hacettepe University Faculty of Medicine Hospital, Department of Anesthesiology and Reanimation, Ankara, Turkey

[2]Hacettepe University Faculty of Medicine Hospital, Department of Cardiovascular Surgery, Ankara, Turkey

List of Abbreviations

ASA: American Society of Anesthesiologists

ASD: Atrial Septal Defect

BIS: Bispectral Index

BMI: Body Mass Index

BSA: Body Surface Area

CI: Confidence Interval

CPB: Cardiopulmonary Bypass

DSA: Density Spectral Array

EEG: Electroencephalography

FiO_2: Fraction of Inspired Oxygen

HFNC: High-Flow Nasal Cannula

ICU: Intensive Care Unit

IQR: Interquartile Range

MICS: Minimally Invasive Cardiac Surgery

MR: Mitral Regurgitation

NIRS: Near-Infrared Spectroscopy

OLV: One-Lung Ventilation

PEEP: Positive End-Expiratory Pressure

STROBE: Strengthening the Reporting of Observational Studies in Epidemiology

TCI: Target-Controlled Infusion

TIVA: Total Intravenous Anesthesia

VSD: Ventricular Septal Defect

Corresponding Author

Name: Dr. Pınar Özdemir Yaşar

Address: Hacettepe University Faculty of Medicine Department of Anesthesiology,
Sıhhiye, Ankara, 06100, TURKEY
Phone: +903123051255
Fax: +903123109600
E-Mail: pozdemiryasar@gmail.com

Abstract

Background: While minimally invasive cardiac surgery has gained growing acceptability in adults, expertise in young patients—particularly regarding anesthetic management—remains scarce. This study evaluated anesthetic methods, perioperative results, and safety profiles in pediatric patients undergoing minimally invasive cardiac surgery at a single tertiary care facility.

Methods: This single-center retrospective cohort study included all pediatric patients aged 5 to 16 years (ASA physical status I to III) who underwent elective minimally invasive cardiac surgery (MICS) through a right anterior mini-thoracotomy for atrial septal defect or mitral regurgitation between 2015 and 2025 (n=8). Anesthesia was delivered with propofol and remifentanil as either total intravenous anesthesia (TIVA, n=5) or target-controlled infusion (TCI, n=3). The primary endpoints were cross-clamp time, time to extubation in the intensive care unit (ICU), ICU and total hospital length of stay, vasoactive-inotropic requirement and 30-day mortality; the secondary endpoint was the occurrence of major postoperative complications. Data were summarized descriptively, proportions are given with exact 95% confidence intervals (CIs), and no comparative hypothesis testing was performed.

Results: Median surgical duration was 210 mins (185–240 mins), anesthesia duration 235 mins (220–260 mins), cardiopulmonary bypass duration 101 mins (91.5–106.5 mins), and cross-clamp duration 63 mins (50–70 mins). Median extubation duration was 150 mins (125–220 mins), median ICU length of stay was 24 hrs (23.5–25.5 hrs), total hospital length of stay was 6.0 [5.0–8.0] days for the TIVA group and 8.0 [6.5–8.5] days for the TCI group, and 30-day mortality was 0% (0/8; 95% CI 0–36.9%). Inotropic support was required in 4 of 8 patients (50%; 95% CI 15.7–84.3%) and successfully discontinued within the first 12 hours postoperatively. The secondary endpoint of postoperative major complications was 0% (95% CI 0–36.9%).

Conclusions: This small single-center retrospective cohort study indicates that pediatric minimally invasive cardiac surgery managed with TIVA or TCI is associated with favorable preliminary perioperative outcomes and stable intraoperative courses in selected patients. Given the exploratory design and limited cohort size, these findings should be interpreted with caution. Nonetheless, this initial experience contributes to expanding clinical literature and underscores the role of multidisciplinary expertise in optimizing patient safety and care.

Keywords: MICS, TIVA, TCI, Pediatric cardiac surgery, Retrospective studies

What is already known?

Minimally invasive cardiac surgery (MICS) in pediatric patients is a domain where advances in perioperative management intersect with evolving anesthesia techniques.

The feasibility of performing MICS in children depends heavily on accurate patient selection and the availability of coordinated expertise from surgical, anesthetic, and critical care teams.

What this article adds

This study confirms that adult MICS protocols are transferable to pediatric care through developmental adjustments and multimodal approaches like TIVA/TCI. It highlights how the synergy between established adult practices and pediatric-specific innovations can yield reproducible positive outcomes and rapid recovery without compromising procedural safety or long-term wellbeing.

INTRODUCTION

Minimally invasive cardiac surgery (MICS) has transformed the treatment of congenital and acquired heart disorders by diminishing surgical trauma, improving aesthetic outcomes, and promoting expedited recovery relative to traditional sternotomy [1, 2]. Although these methods are becoming more prevalent among adults, their use in the pediatric demographic has been constrained, mostly due to technical difficulties, reduced patient size, and safety concerns [3, 4].

Numerous studies have shown that minimally invasive techniques, such as right anterior mini-thoracotomy, axillary thoracotomy, and partial sternotomy, can yield outstanding results in pediatric patients with uncomplicated congenital heart anomalies, including atrial septal defects (ASD) and ventricular septal defects (VSD) [5, 6]. Moreover, longitudinal follow-up data validate the longevity and aesthetic advantages of these procedures [7]. Notwithstanding these advancements in surgical practice, data on anesthetic administration during pediatric minimally invasive cardiac surgery is limited.

Minimally invasive cardiac surgery (MICS) in pediatric patients is a domain where advances in perioperative management intersect with evolving anesthesia techniques. Anesthetic concerns in pediatric minimally invasive cardiac surgery (MICS) are distinctive, encompassing problems related to vascular access, cardiopulmonary bypass techniques, analgesia, and early extubation protocols [4, 5]. Methods like total intravenous anesthesia (TIVA) and target-controlled infusion (TCI) are progressively utilized in adult minimally invasive cardiac surgery (MICS), although their effectiveness and outcomes in pediatric patients remain inadequately defined [3]. The feasibility of performing MICS in children depends heavily on accurate patient selection and the availability of coordinated expertise from surgical, anesthetic, and critical care teams.

This study confirms that adult MICS protocols are transferable to pediatric care through developmental adjustments and multimodal approaches like TIVA/TCI. It highlights how the synergy between established adult practices and pediatric-specific innovations can yield reproducible positive outcomes and rapid recovery without compromising procedural safety or long-term wellbeing. The present retrospective cohort study describes our unit's experience with pediatric patients undergoing MICS with a focus on anesthetic management and perioperative results.

MATERIALS AND METHODS

This retrospective study was conducted in the operating theaters of the Department of Anesthesiology and Reanimation at Hacettepe University between 2015 and 2025. Ethical approval was granted by the Institutional Review Board of Hacettepe University (Approval No: SBA 2025/17-25, Date: 09.09.2025), and written informed consent was obtained from all participating patients or their legal guardians. The study was designed as a single-center retrospective cohort study and is reported in accordance with the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement [8]. Patients were identified by their exposure, namely elective MICS under

one of two predefined intravenous anesthetic techniques, within fixed eligibility criteria and a fixed calendar period, and were then followed through their records for predefined perioperative outcomes; sampling on exposure rather than on outcome defines a cohort rather than a case series, irrespective of cohort size or the presence of a comparison group [9]. The study cohort included pediatric patients aged 5 to 16 years, with an American Society of Anesthesiologists (ASA) physical status of I to III, who were scheduled for elective minimally invasive cardiac surgery. Specifically, the procedures were restricted to right anterior mini-thoracotomy approaches for the surgical correction of Atrial Septal Defect (ASD) and Mitral Regurgitation (MR). Patients requiring emergency surgical interventions were excluded from the study.

Cohort

All pediatric patients who underwent MICS at our institution between 2015 and 2025 and met the eligibility criteria were included consecutively, giving a cohort of 8 patients. We retrospectively collected relevant demographic data (age, weight, BMI, BSA, ASA classification), disease characteristics, intraoperative data (total surgical and anesthetic durations, CPB and cross-clamp times, fluid balance) and postoperative factors, such as vasoactive inotrope scores, rhythm traces, time to extubation, and length of stay in critical care and in hospital from patients. Follow-up to 30 days was complete in all patients.

Anesthetic Technique

Upon arrival in the operating theater, standard cardiac anesthetic monitoring was established for all patients. This included five-lead electrocardiography, pulse oximetry, capnography, invasive arterial blood pressure, central venous pressure (CVP), near-infrared spectroscopy (NIRS), and processed EEG via Bispectral Index (BIS) to assess the depth of anesthesia. External defibrillator pads were positioned prior to draping to facilitate a prompt response in the event of arrhythmias or fibrillation.

General anesthesia was induced utilizing either target-controlled infusion (TCI) or conventional total intravenous anesthesia (TIVA) with propofol and remifentanil. In the TCI group (n=3), induction was achieved using the Eleveld pharmacokinetic model for propofol, set to an effect-site concentration (Ce) of 4.0 µg/mL, and the Minto pharmacokinetic model for remifentanil, set to a Ce of 4.0 ng/mL. In the TIVA group (n=5), induction was initiated with continuous infusions of propofol at a rate of 8 mg/kg/h and remifentanil at 0.05 µg/kg/min. Following the loss of consciousness, neuromuscular blockade was administered using rocuronium (0.6 mg/kg) to facilitate endotracheal intubation, with the depth of the blockade continuously monitored. Following intubation, one-lung ventilation (OLV) was implemented for all patients. Mechanical ventilation parameters included a tidal volume of 5–6 mL/kg, a positive end-expiratory pressure (PEEP) of 5 cmH_2O, and an inspired oxygen fraction (FiO_2) of 0.5–0.8, dynamically adjusted to maintain oxygen saturation >92%. During the maintenance of anesthesia, the administration of both agents was continuously titrated guided by the Density Spectral Array (DSA) trends of the processed EEG to ensure an optimal

depth of anesthesia. In the TCI group, the effect-site concentrations (Ce) were dynamically adjusted using the aforementioned pharmacokinetic models, typically ranging between 3.0 and 4.0 µg/mL for propofol and 2.0 to 4.0 ng/mL for remifentanil, in response to varying surgical stimuli and continuous DSA monitoring. In the TIVA group, adequate anesthesia depth was maintained by titrating continuous infusions of propofol at 6–10 mg/kg/h and remifentanil at 0.1–0.3 µg/kg/min according to the DSA patterns. Effective analgesia was sustained throughout the surgical procedure via the continuous administration of remifentanil in both modalities

All patients underwent a right anterior mini thoracotomy. Peripheral cardiopulmonary bypass (CPB) was established through femoral cannulation. Myocardial protection was attained with antegrade cold blood cardioplegia subsequent to the application of the aortic cross-clamp. Intravenous fluid management consisted of balanced crystalloids at a restrictive rate of 3–5 mL/kg/h, and hemodynamic stability was maintained using boluses of ephedrine (0.1–0.2 mg/kg) or a continuous infusion of norepinephrine (0.02–0.05 µg/kg/min) if the mean arterial pressure decreased by more than 20% from the patient's baseline.

Given the descriptive objective of this retrospective cohort study and its limited sample size (n=8), formal comparative statistical analyses between the TIVA and TCI groups were not performed. Instead, our outcomes were strictly defined as primary and secondary observational endpoints. The primary observational endpoints included cross-clamp time, postoperative extubation time in the intensive care unit (ICU), ICU length of stay, total hospital length of stay, vasoactive-inotropic requirements, and 30-day mortality. The secondary observational endpoint was defined as the occurrence of postoperative major complications.

Study Size

The study size was determined by the total number of eligible patients treated at our institution during the study period, not by an a priori calculation. Because the objective was descriptive and no hypothesis was tested, a formal power calculation was not applicable. In line with the STROBE recommendations, no post hoc power calculation was performed; the precision of the estimates is instead conveyed by 95% CIs [8, 10].

Statistical analysis was performed using SPSS version 25.0 (IBM Corp, 2017). Due to the descriptive design and small cohort size (n=8), comparative analyses (such as hypothesis testing between TIVA and TCI groups) were intentionally omitted. All data are presented using descriptive statistics. Continuous variables are expressed as mean ± standard deviation (SD) or median with interquartile range (IQR), depending on data normality, and categorical variables are reported as frequencies and percentages (%). Proportions are reported with exact (Clopper–Pearson) 95% CIs.

RESULTS

Eight patients underwent MICS 4 of them were male (50 %) and 4 were female (50%). At the time of MICS, the median age and weight were 12 years (9 - 15 years) and 46 kg (37 - 52 kg), respectively. BMI and BSA 20.3 (10.7-

25.1), ; 1.4135 (0.573-1.794) respectively. The diagnoses for surgery were Atrial Septal Defect (ASD) and Mitral Regurgitation (MR). Table 1 summarizes the demographic and clinical characteristics of the patients.

Five patients received TIVA and three received TCI. The median surgical duration was 210 mins (185 –240 mins), anesthesia duration 235 mins (220 – 260 mins), CPB duration 101 mins (91.5 – 106.5 mins), and cross-clamp time 63 mins (50 – 70 mins). Total fluids and urine output 900 (530-1500) and 1000.0 (500.0-1500.0) respectively. The details of intraoperative management are shown in Table 2.

In all patients, the postoperative course was uncomplicated. 4 (50%; 95% CI 15.7–84.3%) patients required inotropic support, with a median vasoactive inotropic score of 1.25 (0.0–5.0), and all inotropes were discontinued by the 12th hour after the operation. There was no evidence of atrial or ventricular arrhythmia in the early postoperative period. We were able to extubate all patients in a median duration of 150 mins (125 - 220 mins), the median length of stay in the ICU was 24 hrs (23.5 – 25.5 hrs), and total hospital length of stay of 6.0 (5.0–8.0) days for the TIVA group and 8.0 (6.5–8.5) days for the TCI group. There was no 30-day mortality and no major complication (0/8 for each; 95% CI 0–36.9%). Postoperative data was depicted in Table 3.

Due to the descriptive design and small cohort size, formal comparative hypothesis testing was not performed; however, individual observations between the TIVA and TCI groups are presented to illustrate our clinical experience. Table 4 shows the results in detail.

DISCUSSION

Eight children underwent right anterior mini thoracotomy for atrial septal defect or mitral regurgitation over a ten-year period. Although a cohort of eight patients cannot test a technique, it can still record what was done and what followed.

Our experience indicates that the anesthetic requirements of adult minimally invasive cardiac surgery, which include one-lung ventilation, peripheral femoral bypass, effect-site targeted intravenous anesthesia, processed EEG and cerebral oximetry, can be met in a group of child patients of median weight 46 kg, and that recovery was not delayed to a degree that would favour sternotomy. While pediatric surgical studies have established the feasibility and durability of this approach [4–7], the specific details of anesthetic management are less frequently reported in the literature.

The duration of recovery in our current study aligns closely with similar timeframes previously recorded in the adult MICS research literature. White and colleagues describe patients extubated within a few hours and an intensive care stay of 24 to 48 hours [11], and the institutional protocol of Demir and Mavioğlu sets early extubation within four to six hours as the target in hemodynamically stable patients [12]. Our median of 150 minutes to extubation and 24 hours

in intensive care is therefore similar with adult ultra-fast-track standards, and the useful observation is that these standards were also met in children.

Jiang et all found that the oxygenation index falls immediately after ultra-fast-track extubation and recovers only by 12 hours, with more frequent recourse to high-flow nasal oxygen [13], so early extubation in a child requires a plan for respiratory support as well as a target time. In line with this, high-flow nasal cannula (HFNC) therapy was employed in our series to provide effective respiratory support whenever postoperative respiratory support was indicated.

Since there are five patients in TIVA and three patients in TCI group, this study cannot support any conclusion comparing the two techniques, and so no comparison was attempted. Total hospital stay was numerically longer in the TCI group (8.0 versus 6.0 days). The pediatric literature addresses this comparison without settling it for cardiac surgery: target-controlled and manual propofol delivery have been compared directly in children [14], as has bispectral index behaviour under the two modes [15], and intravenous maintenance has a long descriptive record in pediatric cardiac anesthesia [16]. Evidence from adults is more substantial. Bai et al. matched elderly cardiac surgical patients receiving target-controlled against manually controlled infusion across two international cohorts and found shorter intensive care stay, lower odds of reintubation, reduced blood pressure variability and a smaller burden of hypotension with TCI [17]. Their work differs from ours in two respects: their infusions ran on the Marsh and Minto models in elderly patients undergoing conventional surgery, whereas we used the latest TCI model “Eleveld” for propofol in children undergoing MICS. Our own method carries a further caveat. The remifentanil component was targeted with a model derived in adults, so its predicted effect-site concentrations in children of this age represent an operational setpoint, since actual exposure was not measured. In both arms the infusion was titrated against the density spectral array of the processed EEG, so the two groups differed in the interface used to reach a target while depth of anesthesia was monitored in the same way.

MICS anesthesia combines cardiac anesthesia with elements of thoracic practice, and most of what it adds concerns the airway [18]. Surgical access through a right anterior mini-thoracotomy requires lung isolation [19, 20], achieved either with a double-lumen tube or with a bronchial blocker whose position is confirmed bronchoscopically [20], and a left-sided double-lumen tube is the conventional choice [13]. In our cohort, lung isolation was successfully achieved in all patients using a left-sided double-lumen tube. External defibrillator pads are positioned before draping because internal cardioversion is often not possible during these procedures [20], and this was our routine in every case. Device dimensions, more than any anesthetic principle, limit how far these techniques transfer to a child. Our cohort, with a median age of 12 years and an interquartile range of 9 to 15, sat in the upper part of the pediatric range where adult-sized equipment becomes practicable.

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oximetry is often where this first becomes apparent [12, 20]. Aquilina et al. documented the sequence during minimally invasive mitral repair: right upper limb saturation fell to 65% after the venous cannula migrated out of the superior vena cava and returned to 100% once it was repositioned [21]. Demir and Mavioğlu give the operational thresholds, a regional saturation below 45 or a fall of more than 20% from baseline, and they place cannula position among the first things to check when these thresholds are breached [12]. The Bonn protocol also includes near-infrared spectroscopy and processed EEG in its standard monitoring bundle [22], and we used these monitors both in all of our patients. Cannulaes are smaller and circulating volüme is lower in a child, so the same degree of displacement can be expected to produce a larger fractional change in venous drainage, which is an argument for continuous cerebral oximetry in this population.

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Eight patients over ten years is roughly one per year. That volume reflects careful case selection, and it also limits what a single centre can demonstrate. These operations are offered to selected children by a team that performs them regularly in adults, and the scientific statement on percutaneous and minimally invasive valve procedures identified the learning curve and cross-specialty team training as determinants of outcome well before these techniques became routine [26]. Case reports continue to show the boundary moving where experience is concentrated, among them a mitral valve replacement through a mini-thoracotomy in a patient with a giant left atrium [27], and the anesthetic skill set has been described as a distinct competency within cardiac practice [28]. Almost all of the literature we have relied on here was written for adults, and until pediatric-specific data accumulate, careful extrapolation remains the only available route and should be identified as such. A multicentre registry recording the endpoints used in this report, time to extubation, intensive care stay, vasoactive requirement and oxygenation after extubation, would allow single-centre experiences of this kind to be pooled.

This small single-center retrospective cohort study indicates that pediatric MICS managed with either TIVA or TCI is associated with favorable preliminary perioperative outcomes in selected patients. Both anesthetic techniques exhibited stable intraoperative courses and uneventful early recovery profiles. Given the exploratory design and limited cohort size of this study, these findings should be interpreted with caution. Nonetheless, our initial experience

contributes to the expanding clinical data on minimally invasive techniques in pediatric cardiac surgery and underscores the value of multidisciplinary expertise in optimizing patient care.

The limitations of this study include its retrospective and observational nature in a tertiary referral center and the small number of subjects. The study size was set by the number of eligible patients and not by an a priori sample size calculation, so the study was not powered to compare the two anesthetic techniques or to detect uncommon complications; the upper 95% confidence limit of 36.9% for zero events among eight patients illustrates how wide the uncertainty around a complication rate of 0% remains. A notable limitation of this study is the specific scope of the surgical procedures analyzed. Although categorized under minimally invasive cardiac surgery (MICS), it is important to acknowledge that MICS encompasses a heterogeneous group of procedures, each presenting unique anesthetic implications and physiological challenges. Because our cohort consisted entirely of patients undergoing right anterior mini-thoracotomy for ASD and MR, our findings regarding anesthetic management and postoperative outcomes may not be entirely generalizable to other MICS approaches, such as mini-sternotomy or robotic-assisted cardiac surgeries. Furthermore, the lack of long-term follow-up data prevents inferences about late outcomes, including neurodevelopment and thoracic growth. The pharmacokinetic models used for target-controlled delivery are a further limitation. Remifentanil was targeted with the Minto model, which was derived in adults, so the effect-site concentrations recorded in the TCI group represent an operational setpoint that has not been validated in children. Further prospective multicenter research may help overcome these limitations.

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CONFLICTS OF INTEREST

None.

COMPETING INTERESTS

The author(s) declare that none

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Table 1. Demographics and Baseline Characteristics

Variable	**Median (IQR) or n (%)**	**Range**
Age (years)	12.0 (9.0–15.0)	5–16
Weight (kg)	46.0 (37.0–52.0)	13–70
ASA class	2.0 (2.0–2.0)	1–3
Gender (1=F,2=M)	2.0 (1.0–2.0)	1–2
BMI	20.3 (18.0–21.0)	10.7–25.1
BSA	1.4135 (1.404-1.423)	0.53-1.794
Surgery Type		
Mitral Valve Replacement(MVR)	5(62.5%)	-
Atrial Septal Defect (ASD)	2(25.0 %)	-
MVR+ Myomectomy	1(12.5%)	-

Table 2. Intraoperative Data

Variable	Median (IQR)	Range
Surgery time (min)	210.0 (185.0–240.0)	135–270
Anesthesia time (min)	235.0 (220.0–260.0)	150–290
CPB duration (min)	101.0 (91.5–106.5)	70–145
Cross-clamp time (min)	63.0 (50.0–70.0)	30–93
Total fluids (ml)	900.0 (530.0–1500.0)	500–3500
Urine output (ml)	1000.0 (500.0–1500.0)	350–6000

Table 3. Postoperative Outcomes

Variable	Median (IQR) or n(%)	Range
Inotropic suport	4 (50%)	
VIS (Vasoactive Inotropic Score)	1.25(0.0-5.0)	0-5
Extubation time (min)	150.0 (125.0–220.0)	120–315
ICU stay (hours)	24.0 (23.5–25.5)	21.5–50
Hospital stay (days)	7.0 (6.0–8.0)	5–9
30-day mortality	0	0
Major complications	None	None

Table 4. Overview of perioperative characteristics and outcomes according to anesthetic technique (TIVA vs. TCI) (Median [IQR])

Variable	TIVA (n=5)	TCI (n=3)
Age (years)	14.0 [9.0–15.0]	14.0 [12.0–15.0]
Weight (kg)	46.0 [37.0–70.0]	47.0 [46.0–52.0]
ASA class	2.0 [2.0–2.0]	2.0 [1.5–2.0]
BMI	20.3 [18.0–23.9]	20.9 [19.0–20.9]
Surgery time (min)	210.0 [200.0–240.0]	210.0 [185.0–217.5]
CPB duration (min)	104.0 [98.0–105.0]	92.0 [81.0–108.5]
Cross-clamp time (min)	60.0 [55.0–69.0]	70.0 [50.0–70.5]
Extubation (min)	150.0 [150.0–155.0]	125.0 [122.5–220.0]
ICU stay (h)	24.0 [23.5–24.5]	25.2 [23.6–25.6]
Hospital stay (d)	6.0 [5.0–8.0]	8.0 [6.5–8.5]

Ranges: Age 5–16 years, Weight 13–70 kg, CPB 70–145 min, Cross-clamp 30–93 min, Extubation 120–315 min, ICU stay 21.5–50 h, Hospital stay 5–9 days.

STROBE Statement

Checklist of items that should be included in reports of observational studies

Section/Topic	Item No	Recommendation	Reported on Page No
Title and abstract	1	(*a*) Indicate the study's design with a commonly used term in the title or the abstract	1
		(*b*) Provide in the abstract an informative and balanced summary of what was done and what was found	2
Introduction			
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported	4
Objectives	3	State specific objectives, including any prespecified hypotheses	4
Methods			
Study design	4	Present key elements of study design early in the paper	4
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	4-5
Participants	6	(*a*) *Cohort study*—Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up *Case-control study*—Give the eligibility criteria, and the sources and methods of case ascertainment and control selection. Give the rationale for the choice of cases and controls *Cross-sectional study*—Give the eligibility criteria, and the sources and methods of selection of participants	4-5
		(*b*) *Cohort study*—For matched studies, give matching criteria and number of exposed and unexposed *Case-control study*—For matched studies, give matching criteria and the number of controls per case	*N/A*
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	5-6
Data sources/measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group	5
Bias	9	Describe any efforts to address potential sources of bias	5, 10
Study size	10	Explain how the study size was arrived at	6
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	6
Statistical methods	12	(*a*) Describe all statistical methods, including those used to control for confounding	6
		(*b*) Describe any methods used to examine subgroups and interactions	6
		(*c*) Explain how missing data were addressed	5
		(*d*) *Cohort study*—If applicable, explain how loss to follow-up was addressed *Case-control study*—If applicable, explain how matching of cases and controls was addressed *Cross-sectional study*—If applicable, describe analytical methods taking account of sampling strategy	5

	(*e*) Describe any sensitivity analyses		N/A

Section/Topic	Item No	Recommendation	Reported on Page No
Results			
Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed	5-6
		(b) Give reasons for non-participation at each stage	5
		(c) Consider use of a flow diagram	N/A
Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders	6, Table 1
		(b) Indicate number of participants with missing data for each variable of interest	6, Tables 1-4
		(c) *Cohort study*—Summarise follow-up time (eg, average and total amount)	5
Outcome data	15*	*Cohort study*—Report numbers of outcome events or summary measures over time	7, Tables 3-4
		Case-control study—Report numbers in each exposure category, or summary measures of exposure	*N/A*
		Cross-sectional study—Report numbers of outcome events or summary measures	*N/A*
Main results	16	(*a*) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included	6-7
		(*b*) Report category boundaries when continuous variables were categorized	N/A
		(*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	N/A
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses	7, Table 4
Discussion			
Key results	18	Summarise key results with reference to study objectives	7
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias	10
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence	7-10
Generalisability	21	Discuss the generalisability (external validity) of the study results	9-10
Other Information			

Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	10

**Give information separately for cases and controls in case-control studies and, if applicable, for exposed and unexposed groups in cohort and cross-sectional studies.*

Note: An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at http://www.plosmedicine.org/, Annals of Internal Medicine at http://www.annals.org/, and Epidemiology at http://www.epidem.com/). Information on the STROBE Initiative is available at www.strobe-statement.org.